DERIVED TENSOR PRODUCTS FOR SYSTEMS OF RINGS AND THEIR FUNCTORIALITY

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Every object of the derived category of systems admits a K-flat representative system. invoked at proof lines 23869-23870. Provides the representative for K.
- A K-flat representative can be used to compute derived tensor products, and the result is independent of the representative on the other side, by the K-flat quasi-isomorphism and derived-tor lemmas. invoked at proof lines 23877-23879. Well-definedness in both variables.
- Exactness of the termwise derived tensor in each variable. invoked at proof lines 23881-23882. Gives the triangulated functoriality.

Source reference. Original lemma 091G: sources/stacks-more-algebra/pinned-original-source.tex, source lines 23816–23816. Cited at proof lines 23870-23870 for the representative of K; the definition and well-definedness are inside the counted span.

Source reference. Original lemma 06Y0: sources/stacks-more-algebra/pinned-original-source.tex, source lines 14192–14192. Cited at proof lines 23878-23878 for well-definedness; the conclusion is inside the counted span.

Source reference. Original lemma 064L: sources/stacks-more-algebra/pinned-original-source.tex, source lines 14457–14457. Cited at proof lines 23878-23879 for well-definedness in the second variable; the conclusion is inside the counted span.

Source reference. Original lemma 064J: sources/stacks-more-algebra/pinned-original-source.tex, source lines 14153–14153. Cited at proof lines 23881-23882 as the model for the exactness argument; the definition and well-definedness are inside the counted span.

Lemma 0.2. *Let (A_n) be an inverse system of rings. Given $K, L \in D(Mod(\mathbf{N}, (A_n)))$ there is a canonical derived tensor product $K \otimes^{\mathbf{L}} L$ in $D(\mathbf{N}, (A_n))$ compatible with the maps to $D(A_n)$. The construction is symmetric in K and L and an exact functor of triangulated categories in each variable.*

Proof. Choose a representative $(K_n^\bullet)$ for K such that each $K_n^\bullet$ is a K-flat complex (Lemma 091G). Then you can define $K \otimes^{\mathbf{L}} L$ as the object represented by the system of complexes

$$(\mathrm{Tot}(K_n^\bullet \otimes_{A_n} L_n^\bullet))$$

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for any choice of representative $(L_n^\bullet)$ for L. This is well defined in both variables by Lemmas 06Y0 and 064L. Compatibility with the map to $D(A_n)$ is clear. Exactness follows exactly as in Lemma 064J. $\square$